

Chapter 17

Magnetic monopoles and superconductors

17.1 Magnetic monopoles

Maxwell's equations can be extended to include magnetic charge alongside electric charge. Denoting the electric charge and current densities by $\rho_e, \vec{j}_e$ and their magnetic counterparts by $\rho_m, \vec{j}_m$, we write

$$\vec{\nabla} \cdot \vec{D} = 4\pi \rho_e \quad , \quad \vec{\nabla} \times \vec{H} = \frac{1}{c} \frac{\partial \vec{D}}{\partial t} + \frac{4\pi}{c} \vec{j}_e \quad (17.1)$$

$$\vec{\nabla} \cdot \vec{B} = 4\pi \rho_m \quad , \quad -\vec{\nabla} \times \vec{E} = \frac{1}{c} \frac{\partial \vec{B}}{\partial t} + \frac{4\pi}{c} \vec{j}_m \quad (17.2)$$

with a continuity equation for each type of charge

$$\vec{\nabla} \cdot \vec{j}_e + \frac{\partial \rho_e}{\partial t} = 0 \quad , \quad \vec{\nabla} \cdot \vec{j}_m + \frac{\partial \rho_m}{\partial t} = 0 \quad (17.3)$$

Let

$$\vec{E} = \vec{E}' \cos \alpha + \vec{H}' \sin \alpha \quad , \quad \vec{H} = -\vec{E}' \sin \alpha + \vec{H}' \cos \alpha \quad (17.4)$$

$$\vec{D} = \vec{D}' \cos \alpha + \vec{B}' \sin \alpha \quad , \quad \vec{B} = -\vec{D}' \sin \alpha + \vec{B}' \cos \alpha \quad (17.5)$$

and correspondingly

$$\rho_e = \rho'_e \cos \alpha + \rho'_m \sin \alpha \quad , \quad \rho_m = -\rho'_e \sin \alpha + \rho'_m \cos \alpha \quad (17.6)$$

$$\vec{j}_e = \vec{j}'_e \cos \alpha + \vec{j}'_m \sin \alpha \quad , \quad \vec{j}_m = -\vec{j}'_e \sin \alpha + \vec{j}'_m \cos \alpha \quad (17.7)$$

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This is a rotation of electric and magnetic charges into one another. It leaves the equations, the energy-momentum tensor $T^{\mu\nu}$, and the force invariant (exercise). We can then choose α to rotate

$$\rho_m \rightarrow 0 \quad , \quad \vec{j}_m \rightarrow 0 \quad (17.8)$$

– assuming all particles have the *same* ratio of electric to magnetic charge. If not, magnetic charge is physically detectable.

Assume then the existence of magnetic charge. A point magnetic charge g sources the field

$$\vec{B} = \frac{g \vec{x}}{|\vec{x}|^3} \quad (17.9)$$

Consider an electric charge e moving past it with speed v at impact parameter b , as shown in the Figure. The transverse component of the magnetic force

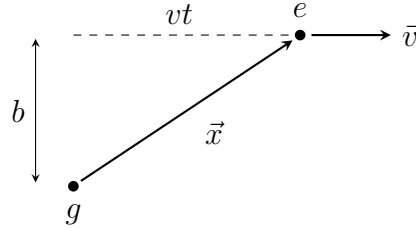


Figure 17.1: A charge e passes a point magnetic charge g with speed v at impact parameter b .

on the charge is

$$F_y = \frac{e g}{c} \frac{v b}{(b^2 + v^2 t^2)^{3/2}} \quad (17.10)$$

so that the net transverse momentum kick is

$$\Delta p_y = \int_{-\infty}^{\infty} F_y dt = \frac{2 e g}{b c} \quad (17.11)$$

The corresponding change in angular momentum, quantized in units of $\hbar$, is

$$\Delta L_z = b \Delta p_y = \frac{2 e g}{c} = n \hbar \quad (17.12)$$

which gives

$$\frac{e g}{\hbar c} = \frac{n}{2} \quad (17.13)$$

$\Rightarrow e$ is quantized. Comparing the two coupling strengths,

$$\alpha_e \equiv \frac{e^2}{\hbar c} \simeq \frac{1}{137} \quad , \quad \alpha_g \equiv \frac{g^2}{\hbar c} \simeq \frac{137}{4} n^2 \gg \alpha_e \quad (17.14)$$

17.2 Superconductors

The London equations are

$$\vec{E} = \frac{\partial}{\partial t} (\Lambda \vec{j}) \quad , \quad \vec{\nabla} \times (\Lambda \vec{j}) = -\vec{B} \quad , \quad \vec{\nabla} \times \vec{H} = \frac{4\pi}{c} \vec{j} \quad (17.15)$$

(with $\Lambda = m/(n q^2)$). Taking the curl of the last equation,

$$\begin{aligned} \vec{\nabla} \times \vec{\nabla} \times \vec{H} &= \frac{4\pi}{c} \vec{\nabla} \times \vec{j} \\ \vec{\nabla} (\vec{\nabla} \cdot \vec{H}) - \vec{\nabla}^2 \vec{H} &= -\frac{4\pi}{\Lambda c} \vec{B} = -\frac{4\pi}{\Lambda c} \vec{H} \end{aligned} \quad (17.16)$$

and since $\vec{\nabla} \cdot \vec{H} = 0$, we get

$$\left(\frac{4\pi}{\Lambda c} - \vec{\nabla}^2 \right) \vec{H} = 0 \quad \Rightarrow \quad \left(\frac{1}{\lambda^2} - \vec{\nabla}^2 \right) \vec{H} = 0 \quad (17.17)$$

defining $1/\lambda^2 \equiv 4\pi/(\Lambda c)$, where λ is the penetration depth.

Consider an infinite slab of thickness $2a$, as depicted in the Figure, in an external field H_0 along z . The solution is

$$H_z(y) = H_0 \frac{\cosh(y/\lambda)}{\cosh(a/\lambda)} [?] \quad , \quad H_z(a) = H_z(-a) \equiv H_0 \quad \text{the external field} \quad (17.18)$$

$\rightarrow$ the superconductor repels the H -field from its interior, as sketched in the Figure.

The origin of this behavior is quantum mechanical. The current is

$$\vec{j} = \frac{i q \hbar}{2m} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) - \frac{q^2}{m c} \vec{A} \psi^* \psi \quad (17.19)$$

where the coupling to the gauge field arises from

$$\vec{\nabla} \rightarrow \vec{\nabla} + i \frac{q}{\hbar c} \vec{A} \quad (17.20)$$

(with $q = 2e$). The last term in (17.19) gives $\vec{j} \propto \vec{A}$

$$\Rightarrow \vec{\nabla} \times \vec{j} \propto \vec{B} \quad (17.21)$$

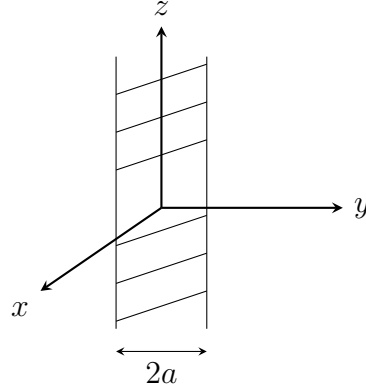


Figure 17.2: An infinite superconducting slab of thickness $2a$, extended along the x and z directions.

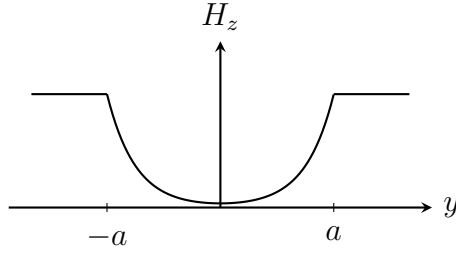


Figure 17.3: The field profile $H_z(y)$ across the slab: the field is expelled from the interior, penetrating only a depth $\sim \lambda$ at the surfaces.

17.3 Flux quantization

The magnetic flux through a loop is

$$\Phi_B = \int \vec{B} \cdot d\vec{a} = \int d\vec{a} \cdot \vec{\nabla} \times \vec{A} = \oint \vec{A} \cdot d\vec{s} \quad (17.22)$$

Take the loop in a region where $\vec{j} = 0$, and write the wavefunction as $\psi = |\psi| e^{i\varphi}$. Then (17.19) gives

$$-\frac{q}{m} |\psi|^2 \left(\hbar \vec{\nabla} \varphi + \frac{q}{c} \vec{A} \right) = 0 \Rightarrow \vec{A} = -\frac{\hbar c}{q} \vec{\nabla} \varphi \quad (17.23)$$

so that

$$\Phi_B = -\frac{\hbar c}{q} \oint \vec{\nabla} \varphi \cdot d\vec{s} = \frac{2\pi n \hbar c}{q} \quad , \quad n \in \mathbb{Z} \quad (17.24)$$

since the phase must return to itself around the loop.